

Patient-Level Cardiac MRI Analysis with Out-of-Fold Heart Segmentation and Shortcut-Learning Controls

A research overview and technical narrative of how 63,425 MRI images became a leakage-controlled 30-patient experiment

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Purpose and research question

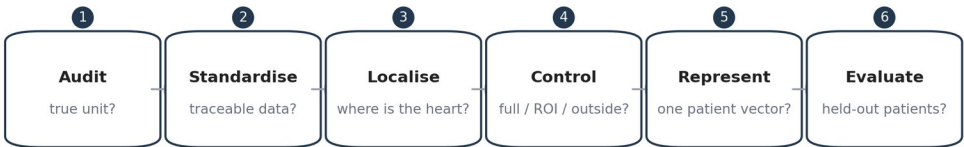
The project asks whether the label-related information in a public cardiac MRI release is concentrated in the cardiac region, or whether a classifier can succeed by exploiting acquisition style, framing, sequence composition, padding, or anatomy outside the heart. The analysis therefore compares three representations of the same source slices: the complete image, an automatically localised cardiac region of interest, and the exact visible-content complement outside that region.

Scope. This is a methodological proof-of-concept study of the released Normal/Sick labels. It is not a clinically validated diagnostic system, does not locate coronary plaques or stenoses, and must not be used for patient care. No source patient images are reproduced; all image-like illustrations in this overview are synthetic.

63,425 JPEG images | **2,859** folder-defined series proxies | **30** patients | **5,097** accepted segmentation targets | **11** feature representations

From a large image archive to a patient-level experiment

The workflow is organised around methodological questions rather than model size.



Central question: is label-related information concentrated in the cardiac region, or can the model rely on shortcuts outside it?

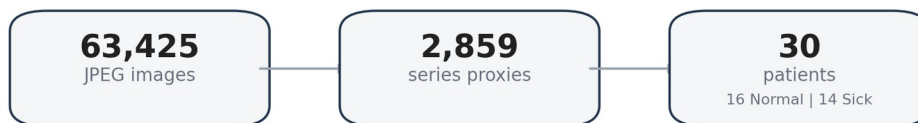
Figure 1. The pipeline is presented as a sequence of methodological questions, from identifying the independent unit to evaluating held-out patients.

1. The dataset audit changed the experiment

At first glance, the release appeared large enough for conventional image classification. A structural audit changed that interpretation. The 63,425 JPEG files were nested beneath only 30 top-level Directory_* folders, and the dataset creator confirmed that each top-level folder represents one patient. The release therefore contained 16 Normal and 14 Sick patient units, with 2,859 immediate or nested folders treated only as series proxies because original DICOM identifiers were unavailable [1,2].

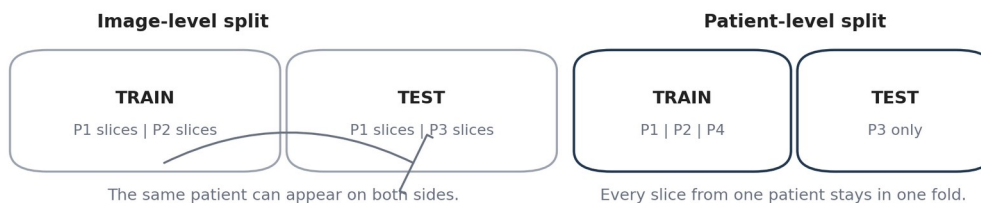
This distinction is decisive. Neighbouring frames from one examination share anatomy, contrast, scanner style, export geometry, and artifacts. If slices from one patient appear in both training and testing, the model may recognise the patient or acquisition pattern instead of learning a transferable disease-related representation. The pipeline therefore assigns folds from patient identity and class label and keeps every descendant image from a patient inside the same fold. Patient identity also governs segmentation, classifier tuning, pooling, bootstrap resampling, and paired comparisons.

The data audit changed the statistical unit



Images enrich the description of a person; they do not create new independent people.

Why image-level splitting can be misleading



Design rule: patient identity governs segmentation folds, classifier folds, tuning, pooling, and uncertainty estimation.

Figure 2. The effective independent sample is 30 patients, not 63,425 images. Patient-level splitting prevents one person from appearing on both sides of an evaluation.

Research consequence. The most important redesign was statistical rather than architectural: the unit of analysis changed from image to patient. Every later result is therefore reported from one out-of-fold score per patient, with uncertainty resampled at patient level.

A duplicate screen found 1,094 exact duplicate groups, all confined within patient and class; no exact cross-patient group was found. A perceptual-hash screen produced 24 cross-patient candidates, including 10 crossing labels, but these remained unconfirmed screening candidates rather than proven duplicates. This unresolved near-duplicate risk is retained as a limitation.

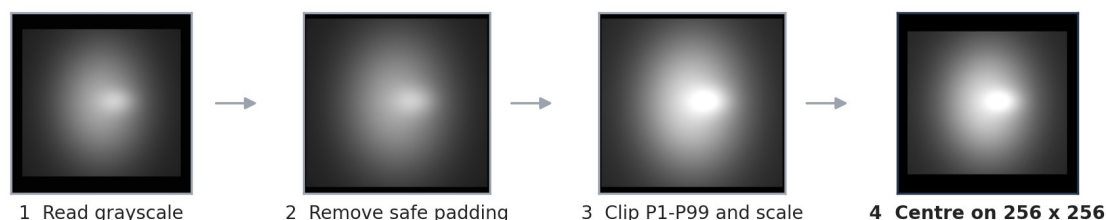
2. Initial CPU stage: establish data contracts before training

The first executable stage builds a manifest that makes every downstream decision traceable. Each image receives a stable token derived from its path beginning at `Directory_*`, together with patient ID, series proxy, sequence position, previous/current/next paths, class label, and patient fold. Stable tokens allow the same image to remain linked to manual masks, quality audits, automatic masks, review decisions, and final predictions even when Kaggle mounts the dataset under a different root.

Preprocessing is deterministic and label-independent. Images are read in grayscale; near-black, low-variance scanner padding is removed conservatively while retaining at least 60% of each original dimension and removing at most 20% from any side. Intensities are clipped to the 1st and 99th percentiles, mapped to $[0,1]$, resized with preserved aspect ratio, and centred on a 256 x 256 canvas. Classification views are later resized to 224 x 224. A content mask distinguishes real image content from newly added canvas padding.

The quality audit measures sharpness, residual-noise ratio, dynamic range, and a perceptual hash. Fixed safeguards are combined with patient-specific distribution tails so that technically poor frames can be marked blurred, noisy, or low-dynamic-range without using the CAD-related label. Manual review then assigns one of three semantic states: **HEART_PRESENT** with a non-empty mask, **NO_HEART_VISIBLE** with an explicit empty target, or **UNUSABLE**. The current audit retained 5,097 training targets across all 30 patients and all five folds: 3,949 heart-present masks and 1,148 no-heart targets. It rejected 161 targets and excluded 1,121 explicitly unusable targets.

Stage 1 - deterministic preprocessing and explicit target semantics



The same geometry is reused for masks; classification views are resized to 224 x 224.

Manual review defines three different meanings

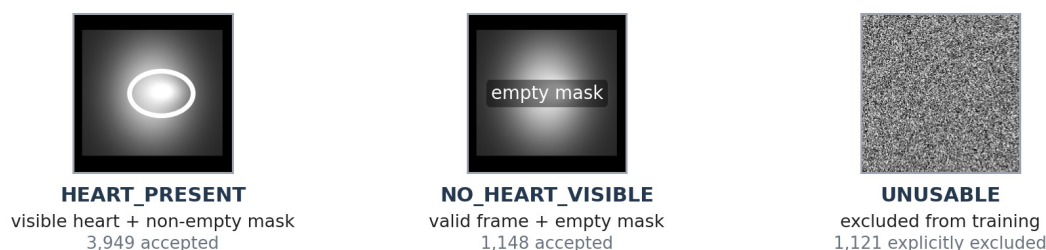


Figure 3. Deterministic preprocessing and the three manual-target meanings. The illustrations are synthetic, not patient images.

Example. For one slice, the manifest fixes its patient and neighbours; preprocessing produces a 256 x 256 robust-scaled view; the reviewer records whether a heart mask should exist; and the accepted target is hashed into the training fingerprint. Changing the label, mask bytes, source, weight, or quality state changes the fingerprint and invalidates dependent checkpoints.

3. GPU stage: train a heart locator, not a disease classifier

The segmentation network never receives the Normal/Sick label. Its task is narrower: propose the general heart region and estimate whether a mask should exist. The model is a 2.5D Attention U-Net [3,4]. Previous, current, and next frames from the same sequence group are stacked as three channels, giving local context without claiming a fully ordered 3D DICOM volume. At sequence boundaries, the nearest available edge frame is repeated.

The encoder uses channel widths 24, 48, 96, and 192 before a 384-channel bottleneck. The decoder restores spatial resolution and uses attention gates to weight encoder skip features with decoder context. The network returns two outputs: a 256 x 256 segmentation logit map and an auxiliary heart-presence logit. The second head helps suppress false-positive masks on localizers or frames in which the heart is not visible.

Training combines 0.40 focal loss, 0.60 Tversky loss, and a 0.30 auxiliary presence binary-cross-entropy term. The Tversky component penalises false-positive mask area more strongly than false negatives (0.65 versus 0.35). Per-target weights preserve distinctions between manual positives, explicit no-heart negatives, and confirmed-auto sources. Data augmentation applies small rotations, scale and translation changes, gamma, contrast, brightness, blur, and noise consistently across all three image channels and the mask. AdamW uses an initial learning rate of 8e-4 and weight decay of 1e-4, for at most 36 epochs with early stopping.

Stage 2 - the segmentation model learns a heart region, not a disease label

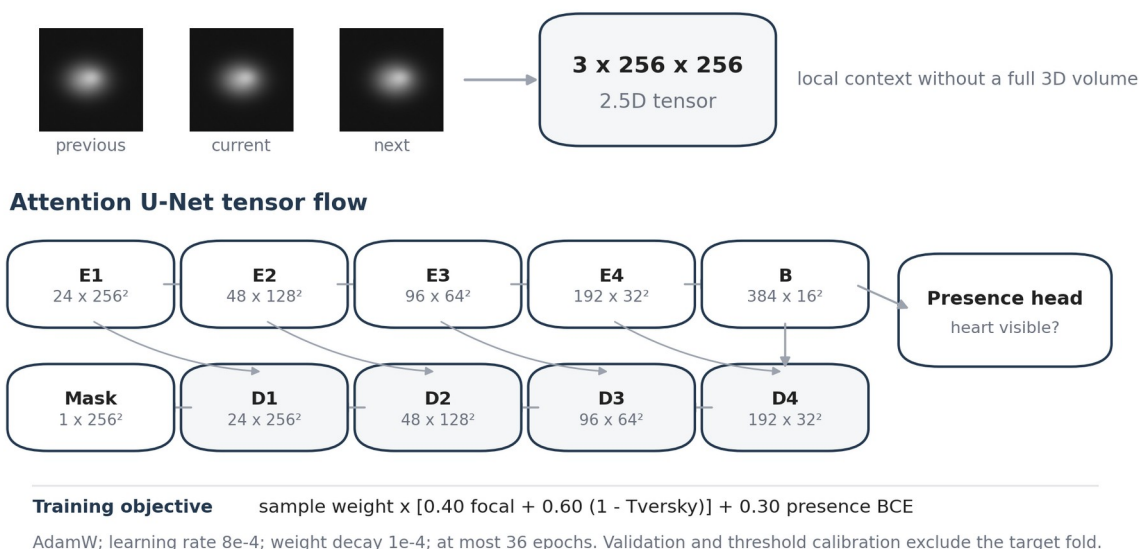


Figure 4. 2.5D input, Attention U-Net tensor flow, presence head, and the combined training objective.

Why 2.5D? A single 2D slice lacks local sequence context, while a true 3D model would require reliable volume ordering, spatial metadata, and more independent patients. The three-frame input is a practical compromise: it adds local context while keeping all convolutions two-dimensional.

4. Out-of-fold prediction turns localisation into a fair downstream input

Five segmentation models are trained at patient level. For target fold k , every patient in that fold is excluded from model fitting and threshold calibration. Validation patients are selected only from the remaining four folds. The held-out fold is predicted once, and the five held-out parts are concatenated. Consequently, every downstream automatic mask comes from a model that did not train on that patient. This is cross-fitting: the segmentation stage itself remains out-of-fold, rather than becoming an unnoticed source of leakage.

The raw mask output is a probability map. A fold-specific threshold creates a binary candidate, a 5 x 5 morphological closing fills small gaps, and only the largest connected component is retained. Plausibility checks then examine mask area, peak probability, boundary contact, and deviation from geometry priors learned from non-test masks. Invalid initial masks are tested at nearby thresholds. The prediction record also stores uncertainty derived from probability entropy, photometric test-time disagreement, heart-presence ambiguity, geometry deviation, and inconsistency with neighbouring frames.

Out-of-fold localisation: each patient is processed by a model that did not learn from that patient



The five held-out parts are concatenated into one complete mask set.

Probability map to final mask



63,107 / 63,425

final-valid automatic masks

0.8088

patient-balanced Dice on heart-present targets

73.4%

accepted no-heart targets predicted empty

Uncertainty combines probability entropy, photometric test-time disagreement, presence ambiguity, geometry deviation, and sequence inconsistency.

Figure 5. Patient-level cross-fitting, probability-to-mask post-processing, and current out-of-fold localisation metrics.

In the complete 5 October 2026 run, 63,107 of 63,425 released images received final-valid automatic masks and 318 remained invalid. Against the 5,097 accepted targets, positive-target patient-balanced Dice was 0.8088. Among explicit no-heart targets, 73.4% received an appropriately empty prediction; approximately one quarter still produced non-empty false-positive masks. These values support useful localisation while documenting a specific failure mode rather than hiding it.

Interpretation boundary. Segmentation quality supports use of the automatic region as an experimental representation. It does not prove anatomical equivalence to expert cardiac contours, because the masks were produced or reviewed by one non-radiologist researcher and were not independently adjudicated by a cardiac radiologist.

5. Region controls test what the classifier can use

The central interpretability test compares three views derived from the same source slice. The full image retains all visible content. The automatic cardiac ROI multiplies the image by a support obtained by expanding the out-of-fold mask with a 15 x 15 maximum-pooling operation. The exact complement retains visible image content outside that same support. Because the three views share the identical source slice, their comparison changes the available region rather than the image-selection distribution.

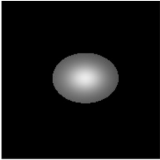
The prespecified primary cohort contains images with valid technical quality, a final-valid out-of-fold mask, positive predicted heart presence, and mask area of at least 0.3% of the image. Exactly 50,735 slices and 2,541 series proxies are used for B2 (full image), AU1 (automatic ROI), and C1 (complement). This same-slice alignment is the strongest comparison in the study.

Three additional cohorts provide context. B0 is the all-slice full-image baseline on 63,425 images. The secondary matched cohort contains 8,875 one-to-one Sick-Normal pairs (17,750 images) selected within acquisition families using appearance, mask geometry, sequence position, and quality; because labels are required to construct pairs, these modes are sensitivity analyses rather than prospective estimates. The manual-positive cohort uses the same 3,949 annotated images to compare manual ROI, automatic ROI, and both complements.

The central control experiment uses different views of the same source slice



Full image
anatomy + background



Automatic cardiac ROI
dilated heart support



Exact complement
visible content outside ROI

Four aligned analysis cohorts

— All released slices	63,425 images B0 full-image baseline
— Primary same-slice cohort	50,735 images B2 full, AU1 ROI, C1 complement
— Secondary matched cohort	17,750 images B1 full, AU2 ROI, C2 complement
— Manual-positive validation cohort	3,949 images M1, AU3, C3, C4

Primary comparisons are strongest because the source slices and patients are identical. Matching and manual-mask analyses are secondary checks.

Figure 6. Full-image, automatic-ROI, and complement controls, followed by the four aligned analysis cohorts and their fixed mode labels.

Negative control logic. If the complement remains predictive, the release still contains label-associated information outside the intended cardiac region. A strong ROI result is therefore encouraging only when interpreted together with the full-image and complement controls.

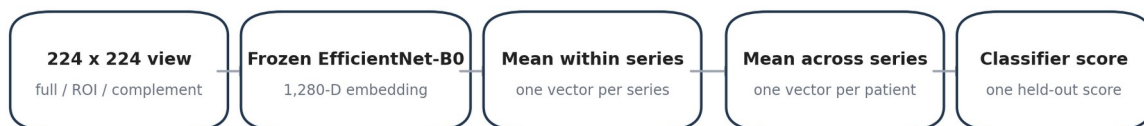
6. Frozen features and hierarchical pooling create one representation per patient

Every experimental view is encoded by the same fixed image model. An ImageNet1K V1-pretrained EfficientNet-B0 converts each 224 x 224 view into a 1,280-dimensional embedding [5]. The network is frozen: no EfficientNet parameter is updated using the CAD-related labels. To preserve the defined numerical protocol, extraction runs on CPU in float32 with autocast disabled, batch size 4, and forward-batch size 4. Eleven feature representations are built across the all-slice, same-slice, matched, and manual-positive cohorts.

Many slices from one patient cannot be treated as independent votes. Slice embeddings are first averaged within each folder-defined series proxy. Series means are then averaged to form one patient vector. This two-level pooling prevents a long series from dominating simply because it contains more exported frames. The final classifier therefore receives one vector per patient, consistent with the independent unit established during the audit.

Classification uses five stratified outer patient folds shared across representations. Inside each outer training set, inner cross-validation selects the logistic-regression regularisation value C from $\{0.01, 0.1, 1, 10\}$ and selects a decision threshold by the Youden index. The fitted pipeline applies standardisation, PCA retaining 95% of variance, and class-balanced logistic regression [6]. Outer-test patients are not used for feature scaling, PCA fitting, hyperparameter selection, or threshold selection.

From many slices to one out-of-fold patient score



Series-first, patient-second aggregation prevents a long scan from acting like many independent people.

Nested patient-level validation



Inside each outer training set

StandardScaler -> PCA retaining 95% variance -> class-balanced logistic regression

Inner cross-validation selects C from $\{0.01, 0.1, 1, 10\}$ and chooses a threshold without examining outer-test patients.

After five folds, patient-level out-of-fold scores are pooled for AUC, AP, Brier score, sensitivity, specificity, and paired bootstrap intervals.

EfficientNet is frozen and runs on CPU/float32 with ImageNet V1 weights and batch/forward size 4.

Figure 7. Frozen EfficientNet feature extraction, series-to-patient pooling, and nested patient-level evaluation.

After all five folds, each patient has one out-of-fold score for each representation. Evaluation reports ROC AUC, average precision, Brier score, accuracy, sensitivity, and specificity. AUC intervals use 2,000 patient-level bootstrap resamples. Representation differences are paired by patient and class and bootstrapped as AUC differences, which is more informative than comparing two unrelated point estimates.

7. Results: cardiac localisation improved ranking, but shortcut signal remained

All values in this section come from the complete 5 October 2026 execution with 5,097 accepted targets and full recomputation of every dependent stage. In the primary same-slice cohort, AU1 achieved AUC 0.938 (95% CI 0.847-1.000), compared with 0.830 for B2 and 0.772 for C1. Paired differences were +0.107 (0.009 to 0.235) versus the full image and +0.165 (0.029 to 0.344) versus the complement.

In the secondary matched cohort, AU2 achieved AUC 0.942, compared with 0.821 for B1 and 0.741 for C2. Paired differences were +0.121 (0.010 to 0.262) and +0.201 (0.054 to 0.384). On the same 3,949 manual-positive images, automatic ROI AU3 achieved AUC 0.973 and manual ROI M1 achieved 0.951; the paired difference M1 minus AU3 was -0.022 (-0.081 to 0.009), so the study does not establish a clear difference or formal equivalence between the two mask sources.

Patient-level results from the complete 5 October 2026 run

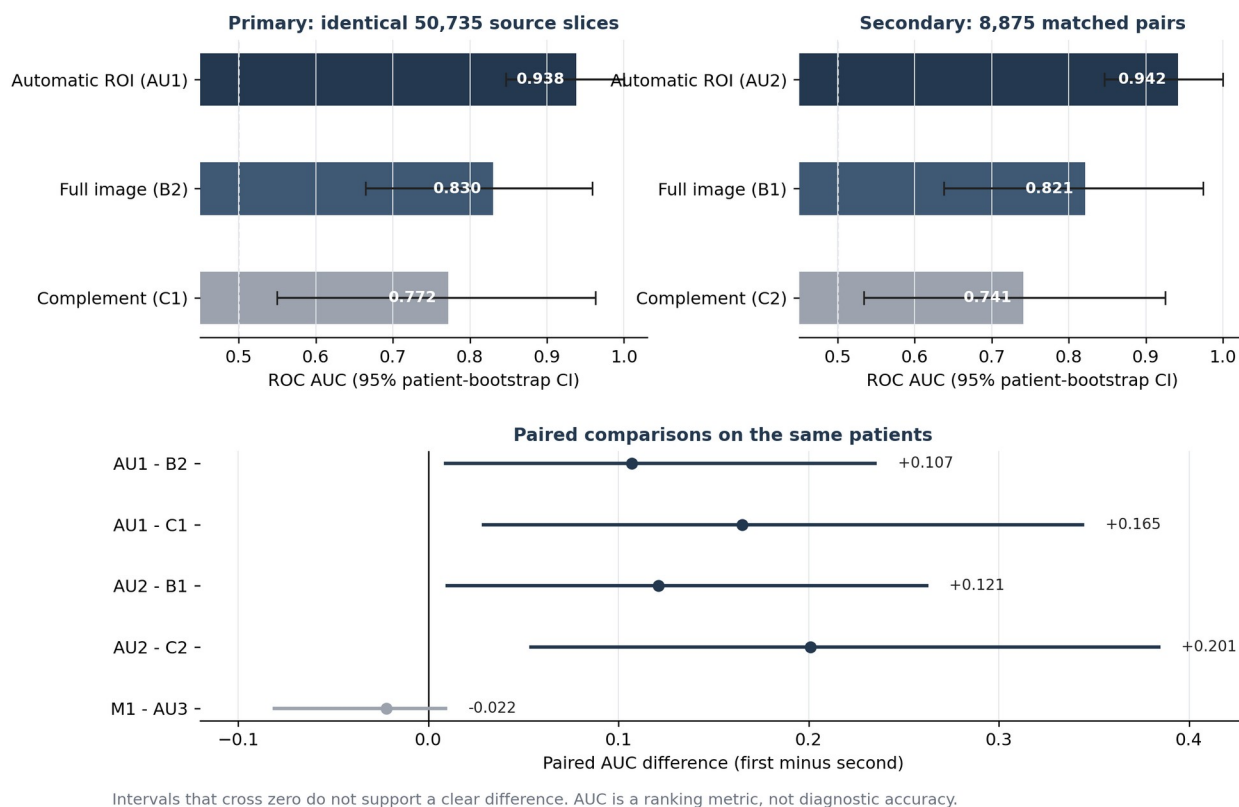


Figure 8. Patient-level AUCs with bootstrap intervals and paired same-patient AUC differences. The primary comparison uses identical source slices.

Supported interpretation. Within this release and protocol, label-related information is more concentrated in the automatically localised cardiac region than in the aligned full image or exact complement. The primary paired intervals for AU1 versus B2 and AU1 versus C1 exclude zero.

Important warning. Complement AUCs from 0.741 to 0.835 remain materially above chance. The model can therefore still exploit extra-cardiac or release-specific information, potentially including acquisition style, sequence distribution, support geometry, surrounding anatomy, JPEG export, or unmeasured confounding. The experiment reduces shortcut explanations; it does not eliminate them.

AUC is not accuracy. An AUC of 0.942 means that patient-level scores rank the supplied labels well across thresholds inside this small release. It does not mean 94.2% diagnostic accuracy and does not establish clinical validity. With only 30 patients, confidence intervals remain wide and sensitive to a few individuals.

8. Reproducibility, contribution, and research responsibility

The implementation is split into three restart-safe Kaggle sessions. The initial CPU stage builds or validates the manifest, image-quality audit, and manual-target audit. The GPU stage trains five fold-specific Attention U-Nets and writes five out-of-fold prediction parts. The final CPU stage performs matching, builds the eleven-mode EfficientNet feature bank, pools patient representations, evaluates classifiers, and reopens the HTML review interface when needed.

Persistence uses two distinct safeguards. Atomic writes first create a temporary PNG, CSV, JSON, PyTorch, or NPZ file and replace the destination only after the new artifact is complete. Semantic fingerprints then decide whether that complete file is logically compatible with current inputs, settings, masks, labels, checkpoints, and cohort structure. An existing filename is never sufficient evidence for reuse. In the reported run, the expanded target state invalidated all five segmentation checkpoints and all five out-of-fold parts; the changed upstream lineage then invalidated and rebuilt all eleven feature representations before evaluation.

Reproducibility architecture and the current level of evidence

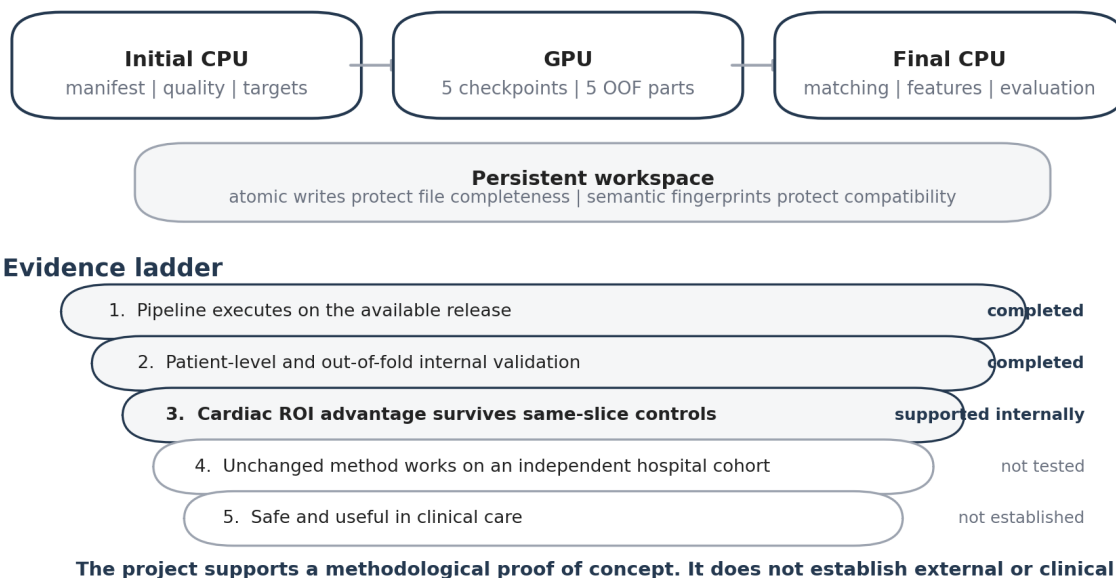


Figure 9. Restart-safe CPU-GPU-CPU execution, the distinction between atomic writing and semantic compatibility, and the current evidence ladder.

Personal contribution and scientific supervision

David Vlad Marghidan conducted the structural audit, contacted the dataset creator to clarify the patient mapping and academic use, redesigned validation around patient-level folds, implemented and refactored the pipeline, reviewed and expanded manual targets, constructed the full/ROI/complement controls, executed the experiments, analysed the results, prepared visualisations, and drafted the research documentation. The work was carried out under the scientific supervision of Prof. Dr. Smaranda Belciug, who provided methodological guidance, challenged assumptions, reviewed conclusions, and provided the INFusion AI research environment.

Use of generative AI

OpenAI ChatGPT was used as an assistive tool for code-review and refactoring discussions, debugging, documentation, language editing, comparison of source versions, consistency checking of author-supplied run outputs, and explanatory figure planning. It did not execute the Kaggle training run and did not generate, estimate, or impute the scientific metrics. Study design, data audit, annotations, execution, validation decisions, interpretation, reference verification, and final responsibility remained human-led under scientific supervision.

9. Limitations, next decisive experiment, and conclusion

Independent sample size. Only 30 patients were available from one public release. Image count cannot compensate for patient scarcity; confidence intervals are wide, and no external cardiac test cohort was evaluated.

Data fidelity. The release contains JPEG exports rather than original DICOM files. Verified sequence identifiers, orientation, spacing, scanner parameters, acquisition metadata, demographics, comorbidities, and site covariates were unavailable.

Outcome and annotations. Folder labels were used as supplied and were not independently re-adjudicated for the 30-patient subset. Manual masks were created or reviewed by one non-radiologist researcher, without independent cardiac-radiologist adjudication or a recorded formal blinding protocol.

Analysis development. Matching uses class labels before classifier cross-validation and is therefore secondary and outcome-informed. Hyperparameters, quality gates, mask-repair rules, and representation definitions were developed on the same release, creating release-specific overfitting risk. No multiplicity correction was applied across exploratory paired comparisons.

Next decisive experiment. Freeze the current protocol and test it unchanged on an independently governed cardiac DICOM cohort with verified patient labels, complete acquisition metadata, radiologist-reviewed masks, site/scanner subgroup analysis, calibration assessment, failure review, and explicit negative controls for padding, support shape, scanner style, and extra-cardiac context.

Conclusion. The most defensible result is dual. Out-of-fold cardiac localisation improved patient-level discrimination relative to exact same-slice full-image and complement controls inside this release. At the same time, materially predictive complements and the very small number of independent patients show that shortcut learning and dataset-specific bias remain important. The study supports a methodological proof of concept, not clinical readiness.

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